

Experiment 2: Water Jug Problem

Aim

To implement the Water Jug problem using Artificial Intelligence search techniques.

Objective

To obtain the required quantity of water using two jugs of different capacities.

Algorithm

1. Start with both jugs empty.
2. Fill one jug completely.
3. Pour water from one jug to the other.
4. Empty a jug whenever it becomes full.
5. Repeat until the required quantity is obtained.
6. Display the sequence of states.

Flowchart

Start

|



Empty Both Jugs

|



Fill a Jug

|

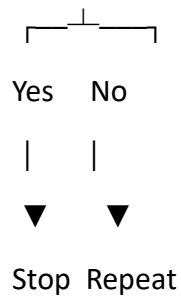


Pour Water

|



Required Quantity?



Python Code

```
from collections import deque
```

```
def water_jug(x, y, target):
```

```
    visited = set()
```

```
    queue = deque([(0, 0)])
```

```
    while queue:
```

```
        a, b = queue.popleft()
```

```
        if (a, b) in visited:
```

```
            continue
```

```
        visited.add((a, b))
```

```
        print((a, b))
```

```
        if a == target or b == target:
```

```
            print("Target reached")
```

```
            return
```

```
        next_states = [
```

```
            (x, b),
```

```
            (a, y),
```

```
(0, b),  
(a, 0),  
(max(0, a-(y-b)), min(y, a+b)),  
(min(x, a+b), max(0, b-(x-a)))  
]
```

```
for state in next_states:  
    if state not in visited:  
        queue.append(state)
```

```
water_jug(4,3,2)
```

Sample Output

```
(0, 0)
```

```
(4, 0)
```

```
(1, 3)
```

```
(1, 0)
```

```
(0, 1)
```

```
(4, 1)
```

```
(2, 3)
```

```
Target reached
```

Result

Thus, the Water Jug problem was solved successfully using state-space search.